



MATERIAL SAFETY DATA SHEET

Section 1. Chemical Product and Company Identification

QINGDAO AUDIS BIO-TECH CO., LTD

CHANGYANG INDUSTRIAL ZONE, LAIXI CITY, QINGDAO, CHINA

Tel: +086 532 58659125 Fax: +086 532 58659126

Product Name: Emamectin Benzoate 5% WDG

Common Name: Emamectin Benzoate

Chemical Name (IUPAC): mixture containing 90% of (10E,14E,16E,22Z)-(1R,4S,5'S,6S,6'R,8R,12S,13S,20R,21R,24S)-6'-[(S)-sec-butyl]-21,24-dihydroxy-5',11,13,22-tetramethyl-2-oxo-3,7,19-trioxatetracyclo[15.6.1.14,8.020,24]pentacosa-10,14,16,22-tetraene-6-spiro-2'-(5',6'-dihydro-2'H-pyran)-12-yl 2,6-dideoxy-3-O-methyl-4-O-(2,4,6-trideoxy-3-O-methyl-4-methylamino-alpha-L-lyxo-hexopyranosyl)-alpha-L-arabinohexopyranoside benzoate and 10% of (10E,14E,16E,22Z)-(1R,4S,5'S,6S,6'R,8R,12S,13S,20R,21R,24S)-21,24-dihydroxy-6'-isopropyl-5',11,13,22-tetramethyl-2-oxo-3,7,19-trioxatetracyclo[15.6.1.14,8.020,24]pentacosa-10,14,16,22-tetraene-6-spiro-2'-(5',6'-dihydro-2'H-pyran)-12-yl 2,6-dideoxy-3-O-methyl-4-O-(2,4,6-trideoxy-3-O-methyl-4-methylamino-alpha-L-lyxo-hexopyranosyl)-alpha-L-arabinohexopyranoside benzoate.

Section 2. Composition/Information on Ingredients

Component CAS	Number	Content (g/kg)
Emamectin benzoate	155569-91-8	5%
Inerts	Not applicable	95%

Section 3. Hazards Identification

2.1 Classification of the substance or mixture

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青岛莱西市昌阳工业园



Classification according to Regulation (EC) No 1272/2008 [EU-GHS/CLP]

Acute toxicity, Oral (Category 4)

Acute aquatic toxicity (Category 1)

Chronic aquatic toxicity (Category 1)

Classification according to EU Directives 67/548/EEC or 1999/45/EC

Harmful if swallowed.

2.2 Label elements

Pictogram

Signal word

Warning

Hazard statement(s)

H302

Harmful if swallowed.

H410

Very toxic to aquatic life with long lasting effects.

Precautionary statement(s)

P273

Avoid release to the environment.

P501

Dispose of contents/ container to an approved waste disposal plant.

Supplemental Hazard Statements

none

Section 4. First Aid Measures

Ingestion:

If swallowed: Call a poison control center or doctor immediately for treatment advice. Have the person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so after calling a poison control center or doctor. Do



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not give anything by mouth to an unconscious person.

Eye Contact:

If in eyes: Hold eye open and rinse slowly and gently with water for 15- 20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

Skin Contact:

If on skin or clothing: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

Inhalation:

If inhaled: Move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth- to- mouth if possible. Call a poison control center or doctor for further treatment advice.

Poisoning Symptoms:

Lack of coordination, Tremors, Dilatation of the pupil

Medical Advice:

This material is believed to enhance GABA activity in animals. It is probably wise to avoid drugs that enhance GABA activity (barbiturates, benzodiazepines, valproic acid) in patients with potentially toxic mectin exposure. Toxicity can be minimized by early administration of chemical absorbents (e.g. activated charcoal). If toxicity from exposure has progressed to cause severe vomiting, the extent of resultant fluid and electrolyte imbalance should be gauged. Appropriate supportive parental fluid replacement therapy should be given, along with other required supportive measures as indicated by clinical signs, symptoms and measurements.



Section 5. Fire Fighting Measures

Fire and Explosion

Flash Point (Test Method): Not Applicable

Autoignition Temperature: Not Available

Flammability: Not Applicable

In Case of Fire

Use dry chemical, foam or CO₂ extinguishing media. Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

Section 6. Accidental Release Measures

In Case of Spill or Leak:

Control the spill at its source. Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of water. Clean up spills immediately, observing precautions in Protective Equipment Section. Cover entire spill with absorbing material and place into compatible disposal container. Scrub area with hard water detergent (e.g. commercial products such as Tide, Joy, Spic and Span). Pick up wash liquid with additional absorbent and place into compatible disposal container. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition.

Section 7. Handling and Storage

Store the material in a well-ventilated, secure area out of reach of children and



domestic animals. Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling. Store in tightly sealed original containers in a dry secure place away from fertilisers, seed, feed and food. Store out of direct sunlight.

Section 8. Exposure Controls/Personal Protection

Ingestion: Prevent eating, drinking, tobacco usage and cosmetic application in areas where there is a potential for exposure to the material. Always wash thoroughly with soap and water after handling.

Eye Contact: Where eye contact is likely, use chemical splash goggles. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

Skin Contact: Where contact is likely, wear chemical-resistant (such as nitrile or butyl) gloves, coveralls, socks and chemical-resistant footwear. For overhead exposure, wear chemical-resistant headgear.

Inhalation: Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below exposure limits. A NIOSH-certified combination air-purifying respirator with an N, P or R 95 or HE class filter and an organic vapor cartridge may be permissible under certain circumstances where airborne concentrations are expected to exceed exposure limits. Protection provided by air-purifying respirators is limited. Use a pressure demand atmosphere-supplying respirator if there is any potential for uncontrolled release, exposure levels are not known, or under any other circumstances where air-purifying respirators may not provide adequate protection.



Section 9. Physical and Chemical Properties

Colour: Pale white

Physical State: Granule

Odor: No characteristic odor

Specific Gravity: 0.6g/cm³

Section 10. Stability and Reactivity

Stability: Stable

Hazardous Polymerization: Will Not Occur

Conditions to Avoid: Heat; light

Materials to Avoid: None known

Hazardous Decomposition Products: Can decompose at high temperatures forming toxic gases.

Section 11. Toxicological Information

Acute Toxicity/Irritation Studies

Oral LD50 (Rat) :1130 mg/kg

Dermal LD50 (Rabbit) : > 2000 mg/kg body weight

Eye Contact: Mild irritant(Rabbit)

Skin Contact: Non-irritant(Rabbit)

Inhalation: > 2.03 mg/L 4 hours

Section 12. Ecological Information

Birds oral > 500mg/kg (Japanese quail, 7d)

Fish 3.05mg/L (Rainbow trout, 96 h)

Daphnia 17.4µg/L (daphnia 48 h)

Bees 0.0614µg/bee (Honeybee 48 h)



Worms > 1000mg/kg(Eisenia foetida dry soil, 14d)

Section 13. Disposal Considerations

Disposal

Do not reuse product containers. Dispose of product containers, waste containers, and residues according to local, state, and federal health and environmental regulations.

Characteristic Waste: Not Applicable

Section 14. Transport Information

Land transport

ADR/ RID:

UN Number: 3077

Transport hazard class: 9

Packing group: III

Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE,
SOLID, N.O.S. (EMAMECTIN BENZOATE 5% WDG)

Environmentally hazardous: Environmentally hazardous

Sea transport

IMDG:

UN Number: 3077

Transport hazard class: 9

Packing group: III

Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE,
SOLID, N.O.S. (EMAMECTIN BENZOATE 5% WDG)

Environmentally hazardous: Environmentally hazardous

Air transport

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IATA-DGR

UN Number: 3077

Transport hazard class: 9

Packing group: III

Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE,
SOLID, N.O.S. (EMAMECTIN BENZOATE 5% WDG)

Environmentally hazardous: Environmentally hazardous

Section 15. Regulator Information

Follow all regulations in your country.

Section 16. Other Information

Disclaimer: The information provided by QINGDAO AUDIS BIO-TECH CO., LTD contained herein is given in good faith and to the best of our knowledge. However, no warranty is expressed or implied.

Prepared by: Register Department

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